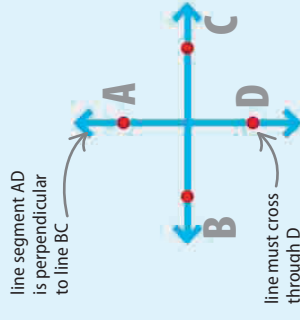
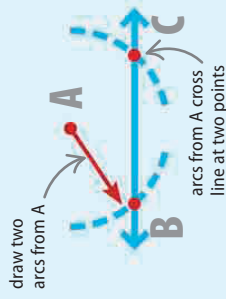
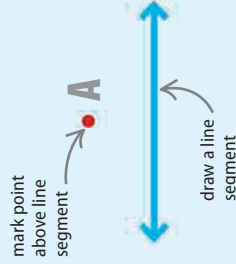


Using a point above the line

Perpendicular lines can be constructed by marking a point above the first line segment, through which the second, perpendicular, one will pass.



Draw a line segment and mark a point above it. Label this point with a letter, for example, A.

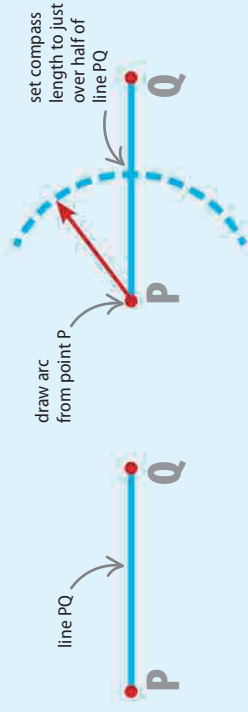
Place a compass on point A. Draw two arcs that intersect the line segment at two points. Label these points B and C.

With the compass on points B and C, draw two arcs of the same length beneath the line segment. Label the intersection of the two arcs point D.

Now, draw a line segment from points A to D. This is perpendicular (at right angles) to line segment BC.

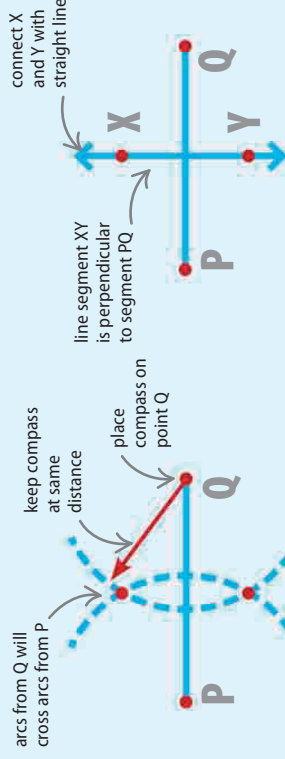
Constructing a perpendicular bisector

A line that passes exactly through the midpoint of a line segment at right angles, or 90° , is called a perpendicular bisector. It can be constructed by marking points above and below the line segment.



First, draw a line segment, and label each end point, for example P and Q.

Place a compass on point P and draw an arc with a distance just over half the length of line segment PQ.



Draw another arc from point Q with the compass kept at the same length. This arc will intersect the first arc at two points.

Label the points where the arcs intersect X and Y. Draw a line connecting X and Y; this is the perpendicular bisector of PQ.